INTERPRETABILITY FAILURE MODES on the VCS

MECHANISM BRANCH	METHOD EXHIBITING IT (phase)	F	*faithfulness F vs intervention oracle (0 … 0.5 … 1.0)*
1 Gradient pathologies *vanishing on the index · saturation · model-invariance* **mechanism: naive position gradient = 0 (§3.2)**	Vanilla saliency	**0.36**	
	SmoothGrad	**0.36**	
	Guided backprop	**0.32**	
	Grad×Input / DeepLIFT	**0.47**	
	Integrated Gradients	**0.36**	
	Expected Gradients	**0.19**	
2 Sampling / approximation error *surrogate variance · baseline sensitivity* **mechanism: IG baseline-sweep → branch 1**	LIME	**0.64**	
	KernelSHAP	**0.65**	
	RISE	**0.54**	
3 Correlational confounds *present ≠ used · spurious tuning · spurious edges* **mechanism: Hewitt & Liang control-task gap**	Tuning curves (A3)	**0.21**	
	Spike-word / pairwise (A4)	**0.25**	
	Granger causality (A6)	**0.16**	
	Linear probing + control (C)	**0.21**	
	Sparse autoencoder (C)	**0.15**	
	NMF/PCA dictionaries (C)	**0.13**	
	Dim-reduction NMF (A7)	**0.52**	
4 Decomposition non-identifiability *polysemantic / rotated atoms ≠ true variables* **mechanism: factorisation atoms not unique**	Single-unit lesions (A2)	**0.96**	
	Attribution patching (C)	**0.48**	
	ACDC circuit disc. (C)	**0.65**	
	Path patching (C)	**0.60**	
5 Compositional blind-spots *single-site misses distributed / interacting structure* **mechanism: 1st-order / greedy-edge approx.**	*Grad-CAM / Grad-CAM++*	**N/A**	*does not apply*
	Attention rollout	**N/A**	*does not apply*
	VIPER	**N/A**	*does not apply*
0 Architecture-specific N/A (control — not a failure of method) *needs conv maps / attention / a learned policy — does not apply* **mechanism: recorded finding (§5) — nothing to score; the subject lacks the substrate**			

(1) gradient pathologies
(2) sampling / approximation
(3) correlational confound
(4) decomposition non-id.
(5) compositional blind-spot
(0) N/A — does not apply (control)
low F (pale chip)
high F (dark chip)
95% bootstrap CI (whisker)